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AIR FORCE-NAVY AERONAUTICAL BULLETIN

SPECIFICATIONS AND STANDARDS;
USE OF

1. Purpose. This bulletin sets forth the order of precedence for the selection of specifications and standard drawings to be used by the Government and Government contractors in the design and construction of aircraft, aeronautical equipment, and other materiel used and procured by the U. S. Air Force and the Bureau of Aeronautics.

2. Order of precedence.

2.1 General.- The requirements of the U. S. Air Force and the Bureau of Aeronautics for materials, processes, and products used in the production of aeronautical material are contained in the applicable standards of the three groups listed herein. Selection of the applicable standards shall be the responsibility of the contractor. In this selection available standards shall be reviewed by these groups in the order of precedence as indicated; for example, if an applicable standard for the required material, process, or product, exists in Group I, it shall be used in preference to any standard in subsequent groups.

2.2 Group I - GOVERNMENT SPECIFICATIONS AND STANDARD DRAWINGS. It is the policy of the U. S. Air Force and the Bureau of Aeronautics to use Government standards wherever such standards are applicable.

2.2.1 Specifications. The applicable specifications are contained in the "Index of Specifications and Related Publications Used by U. S. Air Force, Military Index" for Air Force use, and in the "List of Military Specifications and Standards used by the Bureau of Aeronautics." In reviewing these indexes for an applicable specification, the contractor should review: (a) the Federal specifications listed; if no applicable Federal specification is recorded, (b) the listed Military specifications (including JAN's) shall be reviewed; if no applicable Military specification is recorded, (c) the listed Air Force-Navy Aeronautical (ANA) specifications shall be reviewed; and if no applicable ANA specification is recorded, the other specifications in the index shall be reviewed in the following sequence:

> For Air Force procurement check: (d) applicable U. S. Air Force specifications, and (e) applicable U. S. Army specifications, and (f) other governmental specifications listed in the "Index of Specifications and Related Publications Used by U. S. Air Force, Military Index."

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For Bureau of Aeronautics procurement check: (d) applicable Navy Department specifications, and (e) applicable Bureau of Aeronautics specifications, and (f) other governmental specifications listed in the "List of Military Specifications and Standards used by the Bureau of Aeronautics."

2.2.2 Standard drawings. The applicable standard drawings are contained in the "Numerical Index of U. S. Air Force, Air Force-Navy Aeronautical and Military (MS) Standards" and "Alphabetical List, U. S. Air Force, Air Force-Navy Aeronautical and Military Standard Drawings" for Air Force use, and the "List of Aeronautical Standard Drawings used by the Bureau of Aeronautics" for the Bureau of Aeronautics use. Standard drawings and corresponding standard parts and standard part numbers therefor shall be used wherever they are suitable for the purpose, except those MS or ANA drawings prepared primarily for aircraft powerplant use (six digit numbers).

2.2.3 Where, in the opinion of the contractor, the optimum specification for a specific application is not listed in the appropriate index for that Service, he shall submit an alternate to the procuring activity for approval, selected, if possible, from any of the other indexes in the lists mentioned hereinafter in paragraph 5.3, titled "Indexes of applicable Government standards."

2.3 Group II - ESTABLISHED SPECIFICATIONS AND DRAWINGS OF NON-GOVERNMENT STANDARDIZING ORGANIZATIONS. In the absence of an applicable specification in Group I, the contractor shall review available specifications in Group II, checking first the specifications listed in ANA Bulletin No. 147 which have already been released for use by the Air Force and Bureau of Aeronautics and for which further approval is not required. Specifications and standards released for use in aircraft engines and propellers are defined in ANA Bulletin No. 343 "Specifications and Standards Applicable to Aircraft Engines and Propellers - Use of." The use of a non-Government standard shall be discontinued upon issuance of an applicable Government standard. Standards promulgated by standardizing associations and committees, technical societies, etc, such as American Society for Testing Materials, American Standards Association, National Aircraft Standards Committee, Society of Automotive Engineers (AMS, AS, and ARP series are especially intended for aircraft), Institute of Radio Engineers, and others, are considered a part of Group II. Company standards are not to be considered as a part of this group. Copies of non-Government standards are not available from the Government.

2.4 Group III - COMMERCIAL OR COMPANY SPECIFICATIONS AND DRAWINGS. When no standard exists among those in Groups I and II, a commercial or a company standard may be used, subject to its release by the procuring activity. The use of a commercial or a company standard shall be discontinued upon the issuance of a standard in Group I or Group II covering the same item. For efficient production, it is essential that the number of commercial or company standards used be restricted to the minimum consistent with specific needs.

3. Use of non-Government standards.

3.1 Satisfactory operation of equipment procured under any standards is the responsibility of the contractor, and the use of standards released by the procuring activity is in no way to be considered as a guaranty of the acceptance of the finished product.

3.2 The use of released non-Government standards by the manufacturer will not constitute waiver of Government inspection unless waiver is granted in writing by the procuring activity.

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4. Release by the procuring activity of the Government. The fact that the Government has received drawings or specifications in compliance with a contract shall not be interpreted as release of any standard reference therein unless the standards have been specifically released by the procuring activity of the Government. Justification for the use of the proposed standard, with supporting evidence of its suitability, shall be submitted with a request for release of the standard. Contractors should request release of non-Government materials or products at the time their use is being contemplated.

5. Notes.

5.1 Determination of optimum applicability. When selection of a standard becomes necessary as indicated herein, judgment must be used in weighing technical factors of design and application together with economic factors to make the optimum selection. Among the technical factors to be considered are: safety, performance, strength, weight, production, serviceability, quality control (inspection methods), machinability, corrosion resistance, interchangeability, maintenance and repair, etc. The economic factors include: availability, expense, waste, etc.

5.2 Scope of Government standards.

5.2.1 USAF and Bureau of Aeronautics standards cover materials, processes, and products developed for Air Force or Bureau of Aeronautics use and were generally prepared when need existed for such a standard by only one of the Services. These specifications are no longer being prepared. Specifications known as "limited coordination Military specifications" (with parenthetical designation (USAF), (Aer), or (ASG) included in the specification number) are prepared by the USAF, BuAer, and ASG instead. Specifications of this category are processed into "coordinated Military specifications" in a period of time.

5.2.2 U. S. Army and Navy Department standards cover materials, processes, and products developed especially for the use of more than one of the technical services of the Army or bureaus of the Navy, or the Air Force, as applicable. This category of specifications is no longer being prepared; Military specifications being prepared instead.

5.2.3 Air Force-Navy Aeronautical standards cover materials, processes, and products peculiar to the Air Force and the Bureau of Aeronautics. They were in general developed primarily for use in aircraft structures and such equipment where the prime consideration is reliability from the standpoint of safety, performance, and maximum strength with the minimum weight. This category of specifications is no longer being prepared; Military specifications being prepared instead.

5.2.4 Military specifications cover materials and products, are generally initially prepared by one of the technical services of the Army, Navy, or Air Force and issued as "limited coordination Military specifications," principally for use by that Service only. Within a period of time such specifications are coordinated with all other interested Services of the Army, Navy, and Air Force, and upon approval, are issued as "coordinated Military specifications" for use by all interested Services within the Department of Defense. Both "coordinated" and "limited coordination" Military specifications will be considered as having the same order or precedence insofar as this bulletin is concerned.

5.2.5 Federal specifications are promulgated to cover procurement of interest to Government departments not exclusively the Armed Services and pertain generally to commercial grades. Any of the previously mentioned specifications may be processed into a Federal specification should the proper interest develop among Government services.

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5.3 Indexes of applicable Government standards.

5.3.1 Government specifications approved for Air Force use are listed in the "Index of Specifications and Related Publications Used by U. S. Air Force, Military Index, Volume IV."

5.3.1.1 USAF standard drawings are listed in "Numerical Index of U. S. Air Force, Air Force-Navy Aeronautical and Military Standards" and "Alphabetical List U. S. Air Force, Air Force-Navy Aeronautical and Military Standard Drawings." (Air Force standard drawings are not to be confused with Air Force Production Drawings. Air Force Production Drawings are drawings of Air Force equipment which are not contained in the Standards Book.)

5.3.2 Government specifications approved for Bureau of Aeronautics use are listed in the "List of Military Specifications and Standards Used by the Bureau of Aeronautics."

5.3.2.1 Standard drawings approved for Bureau of Aeronautics use are listed in the "List of Aeronautical Standard Drawings Used by the Bureau of Aeronautics."

5.3.3 All available Federal specifications are listed in the "Index of Federal Specifications and Standards" issued by the General Services Administration.

5.3.4 All available U. S. Army specifications are listed in the "Index of Specifications and Standards Used by Department of the Army, Military Index, Volume II," issued by the Department of the Army.

5.3.5 All Navy Department Military and JAN specifications used by the Navy as a whole are listed in the "Index of Specifications and Standards Used by Department of the Navy, Military Index, Volume III."

5.4 ANA Bulletin No. 147 - Specifications and Standards of Non-Government Organizations. The non-Government specifications and standards listed in ANA Bulletin No. 147 are released for use in the construction of aircraft for the USAF and Bureau of Aeronautics subject to the limitations stated herein.

5.5 Miscellaneous notes on Government specifications and standards (Group I).

5.5.1 Government standards. All drawings prepared by the Air Force and the Bureau of Aeronautics will specify requirements in accordance with Government standards in the order of precedence described herein. Likewise, all specifications prepared by the Air Force and the Bureau of Aeronautics which require the referencing of other specifications and standards will reference Government standards in the order of precedence described herein.

5.5.2 Government purchases. All direct purchases by the Government will be made under applicable Government standards when such are available in the order of precedence listed.

5.5.3 MS drawings. New standard drawing sheets will be numbered in accordance with the Military Standard System. (Example: MS-0000) As existing AN Standard Drawings become inapplicable, MS standards will be issued.

5.6 National Aircraft Standards Committee Standards. This series of non-Government standards is considered part of Group II.

5.7 Society of Automotive Engineers "AMS" series specifications. This series of non-Government specifications is considered part of Group II.

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5.8 Society of Automotive Engineers "AS" and "ARP" series specifications. This series of specifications and standards is considered part of Group II.

5.9 Definitions.

5.9.1 Standard(s). When used as a simple noun in this bulletin, the word shall be interpreted as referring to both specifications and standard drawings.

5.9.2 Release. When used in this bulletin, the word shall be interpreted as a recognition in writing from the Wright Air Development Center, Wright-Patterson Air Force Base, Ohio; or the Bureau of Aeronautics, Navy Department, Washington 25, D. C., or a specifically authorized Air Force or Bureau of Aeronautics representative, as applicable, and making direct reference to the standard by positive means of identification.

5.10 Publications. When requesting publications, refer to both title and number.

5.10.1 Sources. Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

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